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Acute dietary nitrate supplementation improves dry static apnea performance

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ABSTRACT

Acute dietary nitrate (NO_3^-) supplementation has been reported to lower resting blood pressure, reduce the oxygen (O_2) cost of sub-maximal exercise, and improve exercise tolerance. Given the proposed effects of NO_3^- on tissue oxygenation and metabolic rate, it is possible that NO_3^- supplementation might enhance the duration of resting apnea. If so, this might have important applications both in medicine and sport. We investigated the effects of acute NO_3^- supplementation on pre-apnea blood pressure, apneic duration, and the heart rate (HR) and arterial O_2 saturation (SaO_2) responses to sub-maximal and maximal apneas in twelve well-trained apnea divers. Subjects were assigned in a randomized, double blind, crossover design to receive 70 ml of beetroot juice (BR; containing ~ 5.0 mmol of nitrate) and placebo juice (PL; ~ 0.003 mmol of nitrate) treatments. At 2.5 h post-ingestion, the subjects completed a series of two 2-min (sub-maximal) static apneas separated by 3 min of rest, followed by a maximal effort apnea. Relative to PL, BR reduced resting mean arterial pressure by 2% (PL: 86 ± 7 vs. BR: 84 ± 6 mmHg; $P = 0.04$). The mean nadir for SaO_2 after the two sub-maximal apneas was $97.2 \pm 1.6\%$ in PL and $98.5 \pm 0.9\%$ in BR ($P = 0.03$) while the reduction in HR from baseline was not significantly different between PL and BR. Importantly, BR increased maximal apneic duration by 11% (PL: 250 ± 58 vs. BR: 278 ± 64 s; $P = 0.04$). In the longer maximal apneas in BR, the magnitude of the reductions in HR and SaO_2 were greater than in PL ($P \leq 0.05$). The results suggest that acute dietary NO_3^- supplementation may increase apneic duration by reducing metabolic costs.

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1. Introduction

Metabolic rate is a key limiting factor for static apnea performance. Since gas exchange with the environment is interrupted during apnea, oxygen (O_2) used for cellular respiration is mobilized only from the restricted resources in the lung, blood and other tissue (Schagatay, 2009). Therefore, during the course of sustained apnea, the inability to supply O_2 to meet the demand of metabolically active tissue will ultimately result in depletion of O_2 stores and a subsequent accumulation of carbon dioxide (CO_2) and of lactate and hydrogen ions (Schagatay, 2010). Changes in blood and cerebrospinal fluid chemistry, due primarily to increased levels of CO_2 and to some extent diminishing O_2 , are likely the most important physiological factors determining voluntary apneic duration (Feiner et al., 1995; Lanphier and Rahn, 1963; Lin et al., 1974).

Voluntary apneic duration has been reported to be inversely proportional to steady-state O_2 uptake during constant-load

exercise (Astrand, 1960; Craig and Babcock, 1962; Craig and Medd, 1968; Craig and Cummings, 1958). However, O_2 consumption may be reduced during apnea by a cardiovascular diving response, resulting in increased apneic duration (Schagatay and Andersson, 1998). This response, characterized by selective vasoconstriction, bradycardia and reduced cardiac output, may reduce oxygen cost by a combination of greater reliance on anaerobic metabolism in selected tissues, and a reduced myocardial O_2 consumption (Lin, 1982; Lindholm et al., 1999; Schagatay and Andersson, 1998). Beside the diving response, fasting (Lindholm et al., 2007; Schagatay and Lodin-Sundström, 2012) and meditation and Yoga relaxation techniques (Anderson et al., 2000; Liner and Linnarsson, 1994; Telles et al., 2000) appear to be beneficial for apnea performance, while other means of affecting the metabolic rate during apnea have received limited research attention.

Interestingly, several recent studies have shown that dietary nitrate (NO_3^-) supplementation may influence physiological responses both at rest and during exercise. Ingested inorganic NO_3^- is rapidly absorbed by the gut and metabolized to bioactive nitrite (NO_2^-) and further to nitric oxide (NO) (Benjamin et al., 1994; Lundberg and Govoni, 2004). NO is known to be an important regulator of several physiological responses including mitochondrial respiration, mitochondrial biogenesis, calcium homeostasis,

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vascular conductance and blood flow (Cooper and Giulivi, 2007; Dejam et al., 2004; Larsen et al., 2011; Stamler and Meissner, 2001). It is well established that NO_3^- supplementation reduces resting blood pressure (Larsen et al., 2006; Webb et al., 2008). Furthermore, evidence is emerging that both acute and longer-term NO_3^- supplementation may reduce the O_2 cost of exercise and improve exercise tolerance and/or endurance performance in a variety of exercise modalities (Bailey et al., 2009; Lansley et al., 2011a,b; Larsen et al., 2007; Vanhatalo et al., 2010). The question therefore arises: to what extent might the metabolic activity of resting or hypoxic tissue be affected by dietary NO_3^- supplementation?

To our knowledge, no studies have investigated the potential effects of NO_3^- supplementation on apnea performance or the physiological responses to apnea. Since the reduced O_2 cost of exercise following NO_3^- supplementation has been linked to increased muscle efficiency (Bailey et al., 2010), one might expect limited effects during static apnea which is performed with minimal voluntary muscle recruitment (Schagatay, 2009). Moreover, during exercise, the increased O_2 requirement of the active skeletal muscle is met by increased cardiac output, greater muscle perfusion, and increased O_2 extraction (Bangsbo et al., 2000; Korthuis, 2011) whereas, during static apnea, muscle blood flow and O_2 consumption are limited due to a redistribution of the cardiac output to supply mainly the brain and the heart (Schagatay, 2009). Nevertheless, better exercise tolerance and improved oxidative function in hypoxia (Vanhatalo et al., 2011), and a tendency for a reduction in resting O_2 uptake (VO_2) in patients with peripheral arterial disease (Kenjale et al., 2011b), have recently been reported after NO_3^- supplementation. Given these results, the possible effects of NO_3^- supplementation on apnea performance are intriguing.

We therefore investigated whether dietary NO_3^- supplementation could affect static apnea duration and modify the associated physiological responses. Since several physiological responses have been shown to be clearly affected only 2–3 h after NO_3^- intake (Lansley et al., 2011a; Vanhatalo et al., 2010; Webb et al., 2008) we decided to investigate the acute effects of dietary NO_3^- supplementation, using a concentrate of NO_3^- rich organic beetroot juice, on dry static apnea performance. We hypothesized that acute dietary nitrate supplementation would attenuate the fall in SaO_2 during sub-maximal apneas and extend the duration of a maximal apnea.

2. Methods

2.1. Subjects

Twelve healthy subjects (9 men and 3 women) volunteered to participate in this study. Their mean $\pm$ SD age was 32 ± 7 years, height 177 ± 7 cm, weight 69 ± 10 kg, and vital capacity 5.8 ± 1.0 L. All subjects trained for breath-hold diving for at least 2 h/week. One subject smoked regularly, while the remaining were non-smokers. Subjects gave their written informed consent after receiving a description of the procedures and potential risks of participating in the study. The test procedure was carried out in accordance with ethical standards and the Helsinki declaration, which was explained to the subjects. About 1 week before the experiment subjects were asked to adhere to their normal training routine and diet before and between the experimental days. Subjects were instructed to arrive at the laboratory at least 3 h post-prandial, and without consuming caffeine-containing beverages, nicotine or alcohol. Subjects were also told to refrain from exercise or breath-hold activities on the testing days.

2.2. Experimental procedures

Subjects were assigned in a double-blind, randomized, cross-over design to receive dietary supplementation with 70 ml

concentrated NO_3^- rich organic beetroot juice (BR) or placebo (PL) on different days. The BR contained ~ 5.0 mmol of NO_3^- (Beet it, James White Drinks Ltd., Ipswich) and the PL contained ~ 0.003 mmol of NO_3^- (James White Drinks Ltd., Ipswich) (Lansley et al., 2011b). The subjects were instructed to consume the beverages 2.5 h prior to each test. The test days were separated by 1 day, and tests were conducted at the same time of the day (± 2 h) for each subject. Subjects were not given details of the intent of the experiment, and were told that the purpose of the study was to compare the physiological responses during apnea following the consumption of two commercially available juice drinks.

On arrival at the laboratory, the subject's height and body mass were measured, and vital capacity was recorded with a spirometer (Microlab spirometer, Micro Medical Ltd., Kent, UK). All subjects completed apnea testing during both the PL and BR conditions. Apnea series consisted of two 2-min (sub-maximal) apneas separated by 3 min of recovery followed by a 5 min recovery period before a final maximal effort apnea. The series were preceded by 15 min of supine rest, and all apneas and resting periods were performed in supine position on a mattress. Subjects were notified about the time 2 min and 1 min before the apneas, and with 30 s remaining a nose clip was attached. Without prior hyperventilation and after a 10 s countdown period, subjects exhaled fully and performed a deep but not maximal inspiration and started the apnea. This instruction has been shown to result in 85% of VC volume (Schagatay, 1996), which is optimal for dry apneas, and avoids the risk of over filling possibly leading to syncope caused by impeded venous return (Andersson et al., 1998; Schagatay, 2009) and at the same time allows the SaO_2 to fall within a shorter time than should maximal apneas be used. The subjects were given time cues during sub-maximal apneas, and the apneas were terminated on command by countdown from the experimenter. During the maximal apneas, subjects were informed when they had passed 2 min of apneic duration. If SaO_2 levels decreased below 65% during the apneas, the subjects were told to resume breathing in order to avoid hypoxic syncope. After each apneic episode the nose clip was removed. The ambient air temperature was $22 \pm 1^\circ\text{C}$.

2.3. Measurements

Vital capacity was measured in a standing position with a spirometer (Microlab spirometer, Micro Medical Ltd., Kent, UK). Measurements were made in triplicate with the largest value being recorded. HR and SaO_2 were continuously recorded via a combined capnograph–pulse oximeter (Medair Lifesense LS1-9R, Nonin Medical Inc., Medair AB, Delsbo, Sweden), and stored on a memory unit (Trendsense, Nonin Medical Inc., Medair AB, Hudiksvall, Sweden). Probes for registration of HR and SaO_2 were attached to the left forefinger. These variables were recorded from 2 min prior to the first apnea until 2 min after each apnea series. Apneic duration was timed with a stopwatch. Inspired (IV) and expired (EV) lung volumes were measured using a spirometer with the mouthpiece connected to the combined capnograph–pulse oximeter for measurement of end-tidal PCO_2 (ETCO_2). Resting blood pressure was measured twice with an automatic arm cuff system (Omron M41 automated blood pressure monitor, Omron Healthcare, Europe) at the right arm at 12, 9 and 4 min before the apnea, and a mean of the recordings from the last series (4 min before apnea) was used. If blood pressure deviated more than ± 5 mmHg a third measurement was taken.

2.4. Data analysis

All subjects served as their own controls. To determine effects, sub-maximal and maximal apneas with BR and PL were compared for apneic duration and changes in physiological variables. For HR,

the difference between the baseline value (the mean value over a 30 s period starting 1 min before the first apnea) and the value measured over the last 15 s of the apneas were calculated, and the relative difference between the BR and PL conditions was then compared. The degree of SaO_2 was calculated as the percent change at the lowest SaO_2 after the apnea compared to a 30 s period starting 1 min before the first apnea. For the two sub-maximal apneas, mean values were calculated and used for comparison. Mean arterial pressure (MAP) was calculated as $1/3$ of systolic pressure $+2/3$ of diastolic pressure.

2.5. Statistics

The paired Student's *t*-test was used to compare BR and PL values between and within series. Two-way repeated measures ANOVA (mixed model), using GraphPad Prism version 5.04 for Windows, GraphPad Software, La Jolla CA, USA, www.graphpad.com) was used to analyze for differences between PL and BR for the continuous measurement of SaO_2 . Statistical significance was accepted at $P \leq 0.05$, and trends were noted at $P < 0.10$. Data are presented as mean $\pm$ SD for $n = 12$ subjects.

3. Results

BR and PL juices were well tolerated by the subjects. The subjects completed all the designated tests.

3.1. Baseline values

The eupneic resting control values for HR and SaO_2 did not differ between PL and BR, while MAP was $2.3 \pm 3.2\%$ lower in BR ($P = 0.04$; Table 1).

The subjects did not differ in inspired (IV) or expired (EV) lung volume between nitrate and placebo (NS), but had slightly higher end tidal CO_2 after the two sub-maximal apneas in BR compared to PL ($P = 0.01$; Table 1).

3.2. Apneic duration

The mean apneic duration in the maximal apnea test was $11 \pm 12\%$ longer for BR (278 ± 64 s) compared to PL (250 ± 58 s) ($P = 0.04$; Fig. 1a and b).

3.3. Arterial O_2 saturation

The mean SaO_2 at nadir after the two sub-maximal apneas was $97.2 \pm 1.6\%$ in PL and $98.5 \pm 0.9\%$ after BR ($P = 0.03$; Fig. 2).

After the 28 s longer maximal apneas in BR, SaO_2 was lower ($72.7 \pm 9.7\%$) than in PL ($79.6 \pm 9.7\%$; $P = 0.05$). At 155 s into the maximal apnea SaO_2 was $97.0 \pm 2.9\%$ in PL and $97.9 \pm 3.3\%$ in BR (NS), while at 210 s ($n = 9$) into the maximal apnea it was $92.5 \pm 3.9\%$ in PL and $93.0 \pm 5.3\%$ in BR (NS). SaO_2 across the time series was not different between BR and PL either during sub-maximal or maximal apneas (NS, Figs. 3 and 4). No subject reached 65% of SaO_2 during apnea, the safety limit employed for the avoidance of hypoxic syncope

3.4. Heart rate

The typical HR decrease associated with the diving response was seen in all apneas. The decrease from baseline HR during the last 15 s of sub-maximal apneas was $7.2 \pm 14.3\%$ ($P = 0.05$) in PL and $8.2 \pm 19.3\%$ ($P = 0.07$) in BR. The magnitude of the decrease was not significantly different between conditions.

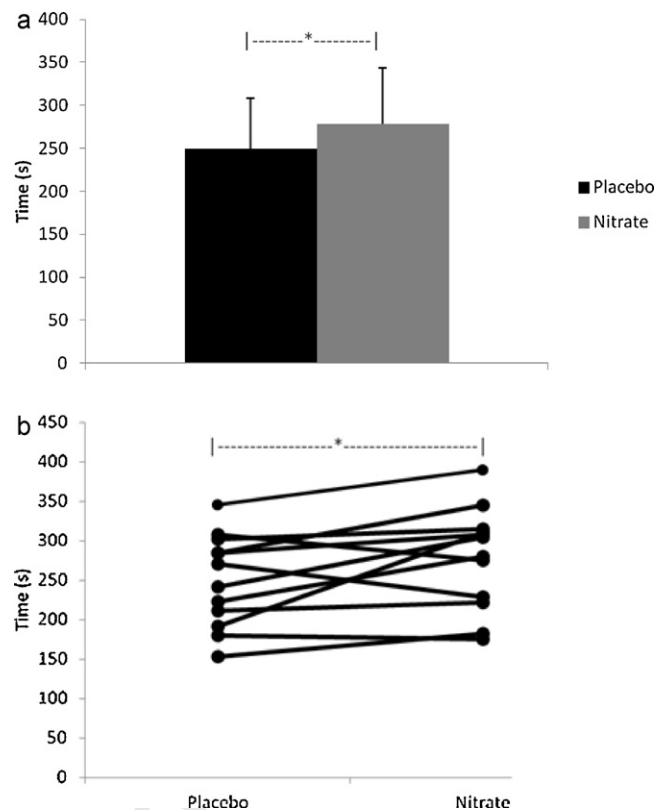


Fig. 1. (a) Mean $\pm$ SD apneic time from maximal apnea during placebo and nitrate trials; $*P \leq 0.05$. (b) Individual apneic time from maximal apnea during placebo and nitrate trials; $*P \leq 0.05$.

The decrease from baseline HR during the last 15 s in the maximal apnea was $17 \pm 21\%$ ($P = 0.02$) in PL and $26 \pm 20\%$ ($P = 0.001$) in BR, with the reduction being greater in BR ($P = 0.03$; Table 1).

4. Discussion

The principal original finding of this investigation was that maximal apneic duration was extended by 11% following acute dietary supplementation with NO_3^- rich BR juice. The mechanism responsible for this effect is unclear but may be due to reduced metabolic rate. This is indirectly supported by the significantly higher SaO_2 observed after BR in the sub-maximal apneas, where apnea duration was the same (2 min) in BR and PL. Since bradycardia was not

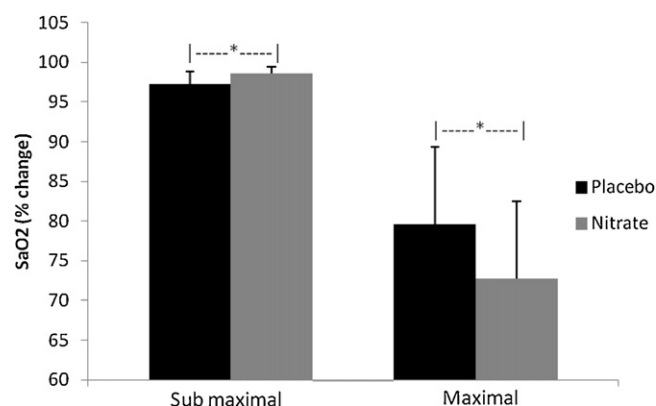


Fig. 2. Mean percent change in nadir arterial haemoglobin oxygen saturation (SaO_2) from the two sub-maximal apneas and maximal apneas in placebo and nitrate conditions. $*P \leq 0.05$. Note that the time to apnea termination was extended following nitrate supplementation.

Table 1
Baseline values for HR, SaO₂ and MAP before apneas during BR and PL trials for 2 min (mean of apneas A1 and A2) and maximal apneas (Amax), and inspired and expired lung volume, end tidal CO₂ during apneas and HR decrease from baseline.

	Mean A1 and A2		A max	
	BR	PL	BR	PL
Baseline MAP (mmHg)	84.3 (5.9)	86.3 ± (7.4)*	–	–
Baseline HR (bpm)	74 ± 15	72 ± 9	77 ± 14	81 ± 16
Baseline SaO ₂ (%)	99.1 ± 0.87	98.9 ± 0.55	99.2 ± 0.8	99.1 ± 0.6
IV (L)	4.4 ± 1.1	4.6 ± 1.0	4.6 ± 0.8	4.5 ± 1.2
EV (L)	4.7 ± 1.0	4.6 ± 1.3	3.6 ± 0.8	3.4 ± 1.0
ET CO ₂ after apneas (%)	6.1 ± 0.5	5.8 ± 0.4**	7.0 ± 1.1	6.8 ± 0.6
HR decrease from baseline (last 15 s of apnea; %)	8.2 ± 19.3	7.2 ± 14.3*	26 ± 20*	17 ± 21*

* P ≤ 0.05.
** P ≤ 0.01.

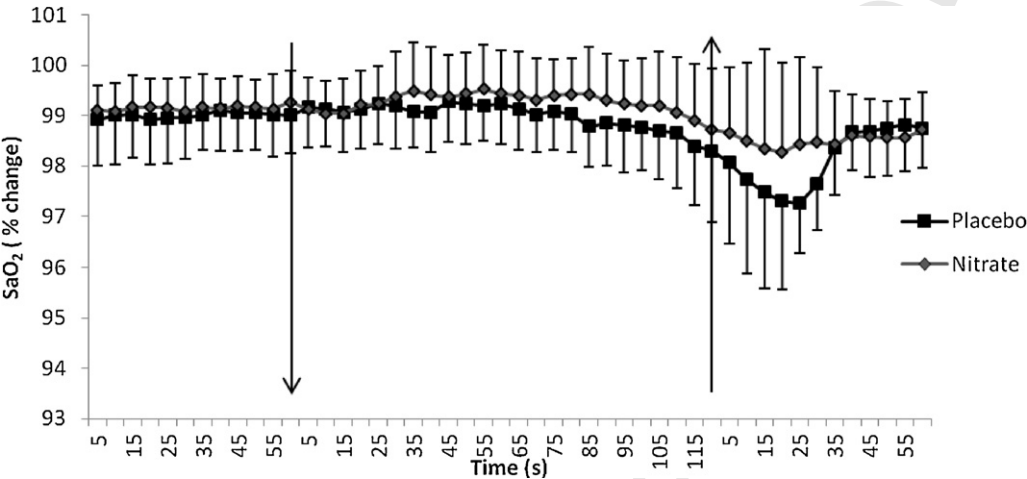


Fig. 3. Mean ± SD percent arterial haemoglobin oxygen saturation (SaO₂) from the two sub-maximal apneas. Arrows indicate start and end of the apneas.

different between BR and PL, the attenuated SaO₂ decrease during BR cannot be explained by metabolic restrictions imposed by this response (Andersson and Schagatay, 1998). It is also unlikely that the pre-apneic gas storage capacity between conditions was different because respiratory parameters were similar between conditions.

The reduced MAP after BR ingestion, despite unchanged HR, suggests that acute nitrate supplementation may have reduced total peripheral resistance at rest. This could possibly counteract the vasoconstriction associated with the diving response during apnea, which is essential for the conservation of O₂. However, the higher SaO₂ during the sub-maximal apneas following BR supplementation implies that O₂ consumption was lower in the BR condition,

possibly reflecting greater efficiency of metabolically active tissue (Larsen et al., 2007, 2011).

After maximal apneas, the SaO₂ nadir was lower after BR ingestion. However, these apneas also lasted nearly half a minute longer, which would predict a greater total O₂ consumption compared to the shorter apneas in PL. It is also likely that the more pronounced HR reduction seen during the maximal apnea with BR resulted from a more established diving response due to the longer apnea duration (Engan et al., 2011). An interesting point is whether the mechanisms regulating voluntary apnea endurance was altered by BR, as the SaO₂ was allowed to fall more before apnea termination. Both hypoxia and hypercapnia are involved in setting the limits of voluntary dives, as both lead to an enhanced respiratory drive

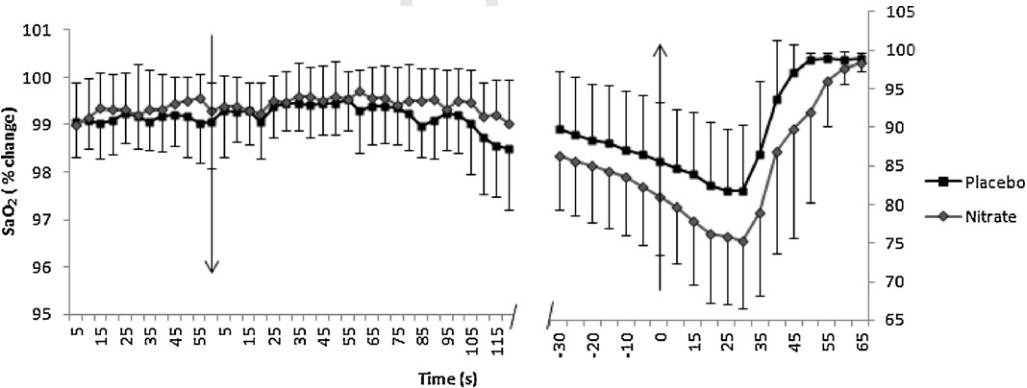


Fig. 4. Mean percent ± SD change in arterial haemoglobin oxygen saturation (SaO₂) from the maximal apneas. Arrows indicate start and end of the apneas. The first 2 min and then the last 30 s of both trials are shown. Note that the time to apnea termination was extended in the nitrate trial.

(Schagatay, 2009). The influence of hypercapnia on the ventilatory response is attenuated in trained apnea divers (Davis et al., 1987; Schaefer, 1955), which makes the input from SaO_2 more important (Lindholm and Lundberg, 2006). The similar ETCO_2 at end-apnea in the present study, however, suggests that the capnic chemoreceptor influence is more important in determining the voluntary apneic duration than the influence of O_2 , which is in accordance with more traditional views of these mechanisms (Lin, 1974). While trained divers may be less sensitive to PCO_2 (Schaefer, 1955), the divers of this study were active but not at their peak of training such that they may have responded more like untrained individuals.

We suggest that the longer maximal apnea duration following NO_3^- supplementation observed in the present study is mainly a result of a lower metabolic rate. While the evidence for this from the present study is indirect and limited, there is support for this possibility in the literature. For example, one previous study reported a strong tendency ($P=0.06$) for reduced O_2 consumption at rest after dietary NO_3^- supplementation in patients suffering from peripheral arterial disease (Kenjale et al., 2011a). In addition, reduced muscle metabolic perturbation and increased exercise tolerance after dietary NO_3^- supplementation has been reported in hypoxia (Vanhatalo et al., 2011). Possible mechanisms for reduced muscle O_2 consumption following NO_3^- supplementation include a reduced ATP cost for a given muscle force production (Bailey et al., 2010) and/or enhanced mitochondrial efficiency (Larsen et al., 2011). It is possible that improved cellular metabolic efficiency and/or mitochondrial efficiency could contribute to a lower resting metabolic rate following NO_3^- supplementation. The longer maximal apnea duration following NO_3^- supplementation might also be related to a more homogeneous delivery of O_2 within tissue thereby reducing CO_2 and H^+ production. Vanhatalo et al. (2010) reported that muscle oxidative function in hypoxia was restored to that measured in normoxia following NO_3^- supplementation and suggested that this might be related to improved perfusion of tissue loci that were relatively more hypoxic. By increasing NO availability, it is possible that NO_3^- supplementation facilitates O_2 distribution in working tissues via partial inhibition of the mitochondria O_2 consumption close to the feed arteriole (Thomas et al., 2001).

The potential physiological effects of NO_3^- might not be consistent across the whole apnea, as systemic hypoxia develops progressively (Andersson and Schagatay, 1998). In fact, deoxyhaemoglobin (Cosby et al., 2003) and deoxymyoglobin (Shiva et al., 2007), which will likely increase due to both peripheral vasoconstriction and increased O_2 consumption relative to limited O_2 delivery during apnea, are important in the reduction of NO_2^- to NO. Since the speed of this reduction is known to occur within one-half circulatory time from artery to vein, this indicates a potential for graded production of NO mainly regulated around haemoglobin oxygenation (Cosby et al., 2003). In addition, hypoxia may also enhance and prolong bioactivity and NO_2^- dependent production of NO in mitochondria (Castello et al., 2006), thus suggesting that O_2 desaturation could potentially increase the amount of NO produced across the apnea. However, it is not known whether the venous constituent of the whole blood, metabolic derivatives, or expression of local tissue factors related to the lack of gas exchange, may influence the involvement, magnitude and timing of these mechanisms during apnea. It should be noted that NO may cause a shift in O_2 -haemoglobin dissociation (Kosaka and Seiyama, 1997; Stepuro and Zinchuk, 2006; Zinchuk, 2003), but it remains unknown whether part of the more pronounced O_2 desaturation during the last part of the maximal effort apneas may be a result of a greater offloading of O_2 from the haemoglobin to the tissue. If this is true, hypoxia or its biochemical consequences could possibly act as a catalyst. In support of this possibility, it is notable that the opposite effect on O_2

saturation was observed during the less desaturated sub-maximal apneas.

Ingestion of the NO_3^- rich BR lowered resting MAP in the healthy normotensive apnea divers, in line with earlier research showing a reduction in systolic and/or diastolic blood pressure after both acute and longer-term NO_3^- administration (Bailey et al., 2009; Larsen et al., 2006; Vanhatalo et al., 2010; Webb et al., 2008). Even though the reduction in MAP was small (2%), this acute reduction was evident 2.5 h after intake without any restriction placed on habitual dietary nitrate and despite being just prior to a test situation. This supports the idea that dietary NO_3^- supplementation may have potential for treatment of pre-hypertension and hypertension (Webb et al., 2008). The mechanistic base for reduced blood pressure after NO_3^- supplementation is suggested to be related to the bioconversion of dietary NO_3^- to NO. NO is known to be a potent vasodilator, and exerts its effects on the vasculature by several mechanisms (Webb et al., 2008). In fact, NO_2^- has also been shown to cause dose-dependent vasodilatation in healthy subjects (Cosby et al., 2003). However, during the course of apnea these effects might, at least theoretically, be counter-productive: whereas the diving response induces a centralized blood shift and reduced perfusion of peripheral tissues (Gooden, 1994; Heistad et al., 1968), NO_3^- supplementation may potentially oppose this by facilitating increased systemic vasodilatation.

Despite the observed effect on SaO_2 during apnea performance, it cannot be ruled out that some physiological changes other than those controlled for in this study contributed to the results. The study did not obtain measurements of pre-apneic resting $\dot{\text{V}}\text{O}_2$, continuous blood pressure or blood flow, which could have been of interest when evaluating the diving response, metabolic rate and blood flow distribution. In addition, we did not measure the plasma NO_3^- or NO_2^- concentration after PL or BR supplementation. However, the dosing protocol used was based on earlier reports which showed that 5 mmol of NO_3^- consumption resulted in a 36% increase in plasma NO_2^- 2.5 h post-ingestion (Vanhatalo et al., 2010). For further research it would be of interest to obtain blood gases in order to investigate potential alterations in PaCO_2 at “breaking point” and thereby central chemoreceptor sensitivity which potentially could permit hypoxia to be tolerated longer.

Based on the present study, dietary NO_3^- supplementation might represent a means of reducing metabolic costs during apnea and increasing apneic duration among trained breath-hold divers. The results also suggest that NO_3^- supplementation might enhance performance in other sports where breathing may be restricted, such as swimming, although additional research is required to investigate this possibility. It would also be of interest to investigate the potential metabolic effects of NO_3^- supplementation during dynamic apnea, as $\dot{\text{V}}\text{O}_2$ is higher and improvement of skeletal muscle efficiency might be expected. Whether our results have implications for clinical conditions which are characterized by tissue hypoxia is unclear. However, the 11% extension of apnea duration and the suggestion that metabolic rate might have been reduced is consistent with the study of Kenjale et al. (2011a). These authors reported that peripheral arterial disease patients were able to walk 18% longer before the onset of claudication pain after dietary nitrate supplementation and 17% longer before terminating exercise, an effect they attributed to increased peripheral tissue oxygenation in areas of hypoxia and possibly reduced metabolic rate. Further research should explore the potential therapeutic and ergogenic effects of NO_3^- supplementation in environmental or clinical conditions which lead to tissue hypoxia or ischaemia.

It should be emphasized that BR was chosen as a source of NO_3^- in the present study because other sources of NO_3^- , such as sodium nitrate and other nitrate salts, may be associated with health risks (Jones et al., 2011; Lundberg et al., 2011). Based on the result of this and other studies, individuals who seek the demonstrated effects

on metabolism and/or blood pressure should be advised to increase their consumption of natural dietary sources rich of NO_3^- such as spinach, beetroot and other green leafy vegetables rather than inorganic sources (Jones et al., 2011; Lundberg et al., 2011).

5. Conclusions

After acute dietary NO_3^- supplementation in trained apneic divers, we found an 11% increase in maximal apneic duration compared to placebo. Sub-maximal apneas of the same duration as in placebo trials resulted in less arterial desaturation after NO_3^- rich BR ingestion, suggesting reduced O_2 consumption being the likely cause of the prolonged maximal apneas. The unaffected diving bradycardia during sub-maximal apneas indicates that NO_3^- supplementation does not affect the cardiovascular diving response. We conclude that dietary NO_3^- supplementation might be a means to enhance apneic performance by reducing metabolic costs for apnea athletes.

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References

- Anderson, J., Larsson, J., Schagatay, E., 2000. Yoga breathing, apneas and alveolar gas exchange. *Undersea and Hyperbaric Medicine* 27.
- Andersson, E.S., Gustafsson, P., Örnhammar, H., 1998. Cardiovascular effects of buccal pumping in breath-hold divers. Annual Scientific Meeting of the European Underwater and Baromedical Society. National Defence Research Establishment, Stockholm, Sweden. FOA Report: FOA-B-98-00342-721-SE: 103-106.
- Andersson, J., Schagatay, E., 1998. Arterial oxygen desaturation during apnea in humans. *Undersea and Hyperbaric Medicine* 25, 21–25.
- Astrand, P.O., 1960. Breath holding during and after muscular exercise. *Journal of Applied Physiology* 15, 220–224.
- Bailey, S.J., Fulford, J., Vanhatalo, A., Winyard, P.G., Blackwell, J.R., DiMenna, F.J., Wilkerson, D.P., Benjamin, N., Jones, A.M., 2010. Dietary nitrate supplementation enhances muscle contractile efficiency during knee-extensor exercise in humans. *Journal of Applied Physiology* 109, 135–148.
- Bailey, S.J., Winyard, P., Vanhatalo, A., Blackwell, J.R., DiMenna, F.J., Wilkerson, D.P., Tarr, J., Benjamin, N., Jones, A.M., 2009. Dietary nitrate supplementation reduces the O_2 cost of low-intensity exercise and enhances tolerance to high-intensity exercise in humans. *Journal of Applied Physiology* 107, 1144–1155.
- Bangsbo, J., Krstrup, P., Gonzalez-Alonso, J., Boushel, R., Saltin, B., 2000. Muscle oxygen kinetics at onset of intense dynamic exercise in humans. *American Journal of Physiology, Regulatory, Integrative and Comparative Physiology* 279, R899–R906.
- Benjamin, N., Odriscoll, F., Dougall, H., Duncan, C., Smith, L., Golden, M., McKenzie, H., 1994. Stomach no synthesis. *Nature* 368, 502.
- Castello, P.R., David, P.S., McClure, T., Crook, Z., Poyton, R.O., 2006. Mitochondrial cytochrome oxidase produces nitric oxide under hypoxic conditions: implications for oxygen sensing and hypoxic signaling in eukaryotes. *Cell Metabolism* 3, 277–287.
- Cooper, C.E., Giulivi, C., 2007. Nitric oxide regulation of mitochondrial oxygen consumption. II: molecular mechanism and tissue physiology. *American Journal of Physiology. Cell Physiology* 292, C1993–C2003.
- Cosby, K., Partovi, K.S., Crawford, J.H., Patel, R.P., Reiter, C.D., Martyr, S., Yang, B.K., Wacławski, M.A., Zalos, G., Xu, X., Huang, K.T., Shields, H., Kim-Shapiro, D.B., Schechter, A.N., Cannon 3rd, R.O., Gladwin, M.T., 2003. Nitrite reduction to nitric oxide by deoxyhemoglobin vasodilates the human circulation. *Nature Medicine* 9, 1498–1505.
- Craig Jr., A.B., Babcock, S.A., 1962. Alveolar CO_2 during breath holding and exercise. *Journal of Applied Physiology* 17, 874–876.
- Craig Jr., A.B., Medd, W.L., 1968. Oxygen consumption and carbon dioxide production during breath-hold diving. *Journal of Applied Physiology* 24, 190–202.
- Craig, F.N., Cummings, E.G., 1958. Breath holding during exercise. *Journal of Applied Physiology* 13, 30–34.
- Davis, F.M., Graves, M.P., Guy, H.J., Prisk, G.K., Tanner, T.E., 1987. Carbon dioxide response and breath-hold times in underwater hockey players. *Undersea Biomedical Research* 14, 527–534.
- Dejam, A., Hunter, C.J., Schechter, A.N., Gladwin, M.T., 2004. Emerging role of nitrite in human biology. *Blood Cells, Molecules, and Diseases* 32, 423–429.

- Engan, H., Richardson, M.X., Lodin-Sundström, A., van Beekvelt, M., Schagatay, E., 2011. Effects of two weeks of daily apnea training on diving response, spleen contraction, and erythropoiesis in novel subjects. *Scandinavian Journal of Medicine & Science in Sports*.
- Feiner, J.R., Bickler, P.E., Severinghaus, J.W., 1995. Hypoxic ventilatory response predicts the extent of maximal breath-holds in man. *Respiration Physiology* 100, 213–222.
- Gooden, B.A., 1994. Mechanism of the human diving response. *Integrative Physiological and Behavioral Science* 29, 6–16.
- Heistad, D.D., Abboud, F.M., Eckstein, J.W., 1968. Vasoconstrictor response to simulated diving in man. *Journal of Applied Physiology* 25, 542–549.
- Jones, A.M., Bailey, S.J., Vanhatalo, A., Fulford, J., Gilchrist, M., Benjamin, N., Winyard, P.G., 2011. Reply to Lundberg, Larsen, and Weitzberg. *Journal of Applied Physiology* 111, 619.
- Kenjale, A.A., Ham, K.L., Stabler, T., Robbins, J.L., Johnson, J.L., Vanbruggen, M., Privette, G., Yim, E., Kraus, W.E., Allen, J.D., 2011a. Dietary nitrate supplementation enhances exercise performance in peripheral arterial disease. *Journal of Applied Physiology* 110, 1582–1591.
- Kenjale, A.A., Ham, K.L., Stabler, T., Robbins, J.L., Johnson, J.L., Vanbruggen, M., Privette, G., Yim, E., Kraus, W.E., Allen, J.D., 2011b. Dietary nitrate supplementation enhances exercise performance in peripheral arterial disease. *Journal of Applied Physiology*.
- Korthuis, R.J., 2011. Skeletal muscle circulation. *Integrated System Physiology: From Molecule to Function to Disease*.
- Kosaka, H., Seiyama, A., 1997. Increased oxygen dissociation by nitric oxide from RBC. *Advances in Experimental Medicine and Biology* 428, 349–354.
- Lanphier, E.H., Rahn, H., 1963. Alveolar Gas Exchange during Breath-Holding Diving. *Techn Docum Rep Amrl-Tdr-63-103 (I)*. AMRL-TR, pp. 79–85.
- Lansley, K.E., Winyard, P.G., Bailey, S.J., Vanhatalo, A., Wilkerson, D.P., Blackwell, J.R., Gilchrist, M., Benjamin, N., Jones, A.M., 2011a. Acute dietary nitrate supplementation improves cycling time trial performance. *Medicine and Science in Sports and Exercise*.
- Lansley, K.E., Winyard, P.G., Fulford, J., Vanhatalo, A., Bailey, S.J., Blackwell, J.R., DiMenna, F.J., Gilchrist, M., Benjamin, N., Jones, A.M., 2011b. Dietary nitrate supplementation reduces the O_2 cost of walking and running: a placebo-controlled study. *Journal of Applied Physiology* 110, 591–600.
- Larsen, F.J., Ekblom, B., Sahlin, K., Lundberg, J.O., Weitzberg, E., 2006. Effects of dietary nitrate on blood pressure in healthy volunteers. *New England Journal of Medicine* 355, 2792–2793.
- Larsen, F.J., Schiffer, T.A., Borniquel, S., Sahlin, K., Ekblom, B., Lundberg, J.O., Weitzberg, E., 2011. Dietary inorganic nitrate improves mitochondrial efficiency in humans. *Cell Metabolism* 13, 149–159.
- Larsen, F.J., Weitzberg, E., Lundberg, J.O., Ekblom, B., 2007. Effects of dietary nitrate on oxygen cost during exercise. *Acta Physiologica (Oxford)* 191, 59–66.
- Lin, Y.C., 1974. Autonomic nervous control of cardiovascular response during diving in the rat. *American Journal of Physiology* 227, 601–605.
- Lin, Y.C., 1982. Breath-hold diving in terrestrial mammals. *Exercise and Sport Sciences Reviews* 10, 270–307.
- Lin, Y.C., Lally, D.A., Moore, T.O., Hong, S.K., 1974. Physiological and conventional breath-hold breaking points. *Journal of Applied Physiology* 37, 291–296.
- Lindholm, P., Conniff, M., Gennser, M., Pendergast, D., Lundgren, C., 2007. Effects of fasting and carbohydrate consumption on voluntary resting apnea duration. *European Journal of Applied Physiology* 100, 417–425.
- Lindholm, P., Lundberg, C., 2006. Alveolar gas composition before and after maximal breath-holds in competitive divers. *Undersea and Hyperbaric Medicine* 33, 463–467.
- Lindholm, P., Sundblad, P., Linnarsson, D., 1999. Oxygen-conserving effects of apnea in exercising men. *Journal of Applied Physiology* 87, 2122–2127.
- Liner, M.H., Linnarsson, D., 1994. Tissue oxygen and carbon dioxide stores and breath-hold diving in humans. *Journal of Applied Physiology* 77, 542–547.
- Lundberg, J.O., Govoni, M., 2004. Inorganic nitrate is a possible source for systemic generation of nitric oxide. *Free Radical Biology and Medicine* 37, 395–400.
- Lundberg, J.O., Larsen, F.J., Weitzberg, E., 2011. Supplementation with nitrate and nitrite salts in exercise: a word of caution. *Journal of Applied Physiology* 111, 616–617.
- Schaefer, K.E., 1955. The role of CO_2 in the physiology of human diving. In: Golf, L.C. (Ed.), *Proceedings of the underwater physiology symposium*. National Academy of Science, National Research Council, Washington, DC, USA, pp. 131–141.
- Schagatay, E., 1996. The Human Diving Response—Effects of Temperature and Training. Lund University, Sweden, Lund.
- Schagatay, E., 2009. Predicting performance in competitive apnoea diving. Part I: static apnoea. *Diving and Hyperbaric Medicine* 39, 88–99.
- Schagatay, E., 2010. Predicting performance in competitive apnea diving. Part 2: dynamic apnea. *Diving and Hyperbaric Medicine* 40, 11–22.
- Schagatay, E., Andersson, J., 1998. Diving response and apneic time in humans. *Undersea and Hyperbaric Medicine* 25, 13–19.
- Schagatay, E., Lodin-Sundström, A., 2012. Fasting Improves Static Apnea Performance in Elite Divers without Enhancing the Risk of Syncope. *Östersund, Sweden*.
- Shiva, S., Huang, Z., Grubina, R., Sun, J., Ringwood, L.A., MacArthur, P.H., Xu, X., Murphy, E., Darley-Usmar, V.M., Gladwin, M.T., 2007. Deoxymyoglobin is a nitrite reductase that generates nitric oxide and regulates mitochondrial respiration. *Circulation Research* 100, 654–661.
- Stamler, J.S., Meissner, G., 2001. Physiology of nitric oxide in skeletal muscle. *Physiological Reviews* 81, 209–237.

- Stepuro, T.L., Zinchuk, V.V., 2006. Nitric oxide effect on the hemoglobin-oxygen affinity. *Journal of Physiology and Pharmacology* 57, 29–38.
- Telles, S., Reddy, S.K., Nagendra, H.R., 2000. Oxygen consumption and respiration following two yoga relaxation techniques. *Applied Psychophysiology and Biofeedback* 25, 221–227.
- Thomas, D.D., Liu, X., Kantrow, S.P., Lancaster Jr., J.R., 2001. The biological lifetime of nitric oxide: implications for the perivascular dynamics of NO and O₂. *Proceedings of the National Academy of Sciences of the United States of America* 98, 355–360.
- Vanhatalo, A., Bailey, S.J., Blackwell, J.R., DiMenna, F.J., Pavey, T.G., Wilkerson, D.P., Benjamin, N., Winyard, P.G., Jones, A.M., 2010. Acute and chronic effects of dietary nitrate supplementation on blood pressure and the physiological responses to moderate-intensity and incremental exercise. *American Journal of Physiology. Regulatory, Integrative and Comparative Physiology* 299, R1121–R1131.
- Vanhatalo, A., Fulford, J., Bailey, S.J., Blackwell, J.R., Winyard, P.G., Jones, A.M., 2011. Dietary nitrate reduces muscle metabolic perturbation and improves exercise tolerance in hypoxia. *Journal of Physiology*.
- Webb, A.J., Patel, N., Loukogeorgakis, S., Okorie, M., Aboud, Z., Misra, S., Rashid, R., Miall, P., Deanfield, J., Benjamin, N., MacAllister, R., Hobbs, A.J., Ahluwalia, A., 2008. Acute blood pressure lowering, vasoprotective, and antiplatelet properties of dietary nitrate via bioconversion to nitrite. *Hypertension* 51, 784–790.
- Zinchuk, V.V., 2003. The involvement of nitric oxide in formation of hemoglobin oxygen-binding properties. *Uspekhi Fiziologicheskikh Nauk* 34, 33–45.